Appendix: Deferred Proofs, Implementation Details, and Further Results

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1 Proofs of Section 2 of the main text

Statements are restated for reference; equation numbers refer to the main text.

1.1 Channel 1: the dead knob

Proposition 1.1 (Dead knob, following Caveat 1 of Kolchinsky and Tracey [2019])**.** *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X;Z|Y) - \gamma I(Y;Z)$ is minimized exactly on the corner family $\{Z : I(Y;Z) = H(Y),\ I(X;Z|Y) = 0\}$, and the map $\gamma \mapsto \big(I(X;Z^*|Y),\ I(Y;Z^*)\big)$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

Proof sketch (see Kolchinsky and Tracey [2019] for the full argument). By the chain rule, $L_\gamma(Z) = I(X;Z) - (1+\gamma)I(Y;Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper's maximizing convention this is their Eq. 7 with $\beta' = 1/(1+\gamma) \in (0,1)$, the range in which every maximizer is pinned at the copy corner. □

1.2 Channel 2: the blind certificate

Proposition 1.2 (The certificate records nothing about inter-label geometry)**.** *For any* fixed *encoder, with the backward encoder matched to the posterior per class ($b = q(z|y)$), the certificate collapses to the residual,* $\mathrm{ReX} = I(X;Z|Y)$*, whatever the pairwise geometry of the class-conditional priors $\{q(z|y)\}$: class means merged, coincident, or placed in arbitrary directions all yield the same certificate value. Formally, the* certificate *contains no cross-class term — it is a sum of per-class matching terms, each sample's KL touches only its own class prior, and the matching condition can be met class by class regardless of the geometry of $\{q(z|y)\}$ — so the certificate never records the relationships among the class priors.*

Proof. For any matched pair ($b = q(z|y)$), the gap identity of Eq. (2) of the main text gives $\mathrm{ReX} = I(X;Z|Y)$; the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of $\{q(z|y)\}$. Note that $I(X; Z|Y)$ is generally non-zero; it vanishes only under the conditional independence $Z \perp X \mid Y$, which is a property of the encoder, not of the prior. $\square$

Proposition 1.3 (Retention blindness on T_α)**.** *On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky and Tracey [2019] (their Eq. 6) — a family that depends on Y alone — $I(X; T_\alpha|Y) = 0$ for every α, and with the optimal backward encoder and classifier, $\mathrm{ReX}(T_\alpha) = 0$ and $\mathrm{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha$: the certificate reads 0 from the total collapse ($\alpha = 0$) to the full copy ($\alpha = 1$), while the retained label information $I(Y; T_\alpha)$ sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of Y — the certificate records nothing about how much label information the code retains.*

Proof. T_α is a function of Y alone, hence $T_\alpha \perp X \mid Y$ and $I(X; T_\alpha|Y) = 0$. By Proposition 1.2 with the true backward encoder the certificate is tight, $\mathrm{ReX} = I(X; T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$, i.e. $\mathrm{CE} = H(Y|T_\alpha) = \delta_\alpha$. $\square$

Remark 1.1 (The objective on T_α is β-independent and strictly prefers the full copy). On T_α the compression term contributes nothing ($\mathrm{ReX} \equiv 0$), so the objective reads $L = \mathrm{CE} + \beta\,\mathrm{ReX} = \delta_\alpha$ — independent of β, and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits. Any statement about trading label information against certified residual would require a path on which $\mathrm{ReX} > 0$; on T_α no such trade occurs.

1.3 The GPB framework: certificate preservation and separation transfer

Proposition 1.4 (A stateless geometric prior preserves the CEB certificate)**.** *For any prior family $\mathcal{G}$ whose members are conditional priors independent of x, and for every parameter value θ at which $p_\theta \in \mathcal{G}$ is defined, the gap identity applies pointwise in θ:*

$$\mathbb{E}\Big[\mathrm{KL}\big(q_\phi(z|x) \,\|\, p_\theta(z|y)\big)\Big] = I_{q_\phi}(X; Z|Y) + \mathbb{E}_Y\Big[\mathrm{KL}\big(q_\phi(z|Y) \,\|\, p_\theta(z|Y)\big)\Big] \;\geq\; I_{q_\phi}(X; Z|Y). \tag{1}$$

The recomputation of p_θ after parameter updates does not invalidate this identity: at each fixed θ it remains a function of y only. If the prior were computed from the current input batch, this conditional-prior argument would no longer apply without a new analysis.

Proof. For fixed θ, apply the gap identity of the main text to the conditional distribution $p_\theta(z|y)$ and the posterior q_ϕ. $\square$

Proposition 1.5 (Separation transfer for any qualified geometric prior)**.** *Let $\mathcal{G}$ be qualified with variance bound $\tau^2_{\max}$.* (a) Discrete labels. *Let $m_k := \mathbb{E}[\mu_\phi(X)|Y =$*

$k]$ be the posterior class centers and define the class-wise anchor mismatch $D_k := \frac{1}{2\tau_{\max}^2}\mathbb{E}\big[\|\mu_\phi(X) - m_\theta(k)\|^2 \mid Y = k\big]$. Then for every pair $i \neq j$,

$$\|m_i - m_j\| \;\geq\; \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j} \;\geq\; \Delta - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j}. \tag{2}$$

(b) Continuous labels. *With $m(y) := \mathbb{E}[\mu_\phi(X) \mid Y = y]$ the regular conditional posterior center and $D(y) := \frac{1}{2\tau_{\max}^2}\mathbb{E}\big[\|\mu_\phi(X) - m_\theta(y)\|^2 \mid Y = y\big]$, for P_Y-almost every pair $y \neq y'$,*

$$\|m(y) - m(y')\| \;\geq\; \|m_\theta(y) - m_\theta(y')\| - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')} \;\geq\; \rho_{\min}\, d_Y(y, y') - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D} \tag{3}$$

no global positive minimum separation being required. Small pointwise anchor mismatch therefore guarantees that the posterior centers inherit the prior's separation for any *qualified member of $\mathcal{G}$.*

Proof. For (a), Jensen gives $\|m_k - m_\theta(k)\| \leq \sqrt{2\tau_{\max}^2 D_k}$, and the triangle inequality on $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$ gives the bound. For (b), the same two inequalities hold pointwise for P_Y-a.e. y: Jensen on the regular conditional expectation gives $\|m(y) - m_\theta(y)\| \leq \sqrt{2\tau_{\max}^2 D(y)}$ (the norm is convex), and the triangle inequality gives the bound; intersecting the two measure-one sets of valid y's yields the almost-everywhere pair statement. $\square$

Remark 1.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max}\,\mathrm{diam}(B_i) + \rho_{\max}\,\mathrm{diam}(B_j)$, which vanishes as the bins shrink. The theorem itself is stated pointwise and needs no binning.

Proposition 1.6 (Anchor alignment at near-optima: deterministic K-class classification)**.** *Assume the deterministic K-class classification setting: $Y = f(X)$, a cross-entropy prediction loss, and a qualified family whose diagonal covariances belong to the diagonal family the posterior can represent. Suppose further that the encoder family contains class-constant maps, so that the aligned solution $q(z|x) = p_\theta(z|y)$ is feasible, and that the classifier family contains the Bayes posterior of the aligned mixture,*

$$c_k(z) = \frac{\pi_k\, |\Sigma_\theta(k)|^{-1/2}\, e^{-\frac{1}{2}(z - m_\theta(k))^\top \Sigma_\theta(k)^{-1}(z - m_\theta(k))}}{\sum_j \pi_j\, |\Sigma_\theta(j)|^{-1/2}\, e^{-\frac{1}{2}(z - m_\theta(j))^\top \Sigma_\theta(j)^{-1}(z - m_\theta(j))}}, \tag{4}$$

including the Gaussian normalization factors $|\Sigma_\theta(k)|^{-1/2}$. For the orthogonal instance, where all classes share the covariance $\tau^2 I_d$ and the means are the anchors $a\, q_{\theta,k}$, the determinant factors cancel and this Bayes posterior reduces to exactly the anchor-energy classifier, whose weights $w_k = (a/\tau^2)\, q_k$ are fixed by construction. Then for any solution (ϕ, θ, ψ) with objective value $L \leq L^* + \varepsilon$, where L^* is the *infimum of the main-text objective, the expected anchor mismatch is controlled by β:*

$$\mathbb{E}\big[D(Y)\big] \;\le\; \frac{C_{\mathrm{G}}+\varepsilon}{\beta} \;\le\; \frac{\ln K+\varepsilon}{\beta}, \qquad D(y) := \frac{1}{2\tau_{\max}^2}\,\mathbb{E}\Big[\|\mu_\phi(X)-m_\theta(y)\|^2 \;\Big|\; Y=y\Big], \tag{5}$$

where $C_{\mathrm{G}} := H(Y \mid Z_{\mathrm{align}})$ is the cross-entropy of the Bayes classifier on the aligned mixture, bounded by $C_{\mathrm{G}} \le H(Y) \le \ln K$; in the shared-covariance orthogonal instance it specializes to the anchor-energy classifier's aligned cross-entropy C_{OPB}. For tasks in which no zero-KL feasible solution exists — in particular for continuous regression, where the encoder cannot recover y from x — the argument does not apply: there the separation transfer remains a conditional statement on the anchor mismatch.

Proof. Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$, each sample's KL contains a mean-mismatch term with variance at most $\tau_{\max}^2$, so $\mathrm{KL}(q_\phi(z|x)\,\|\,p_\theta(z|y)) \ge \|\mu_\phi(x)-m_\theta(y)\|^2/(2\tau_{\max}^2)$, and taking expectations gives $\mathbb{E}[\mathrm{KL}] \ge \mathbb{E}[D(Y)]$. The comparison solution $q(z|x) = p_\theta(z|y)$ is feasible by the assumptions ($Y = f(X)$, class-constant maps, diagonal representable covariances), has zero KL, and its prediction is the Bayes posterior, whose cross-entropy is exactly the conditional entropy $C_{\mathrm{G}} = H(Y \mid Z_{\mathrm{align}}) \le \ln K$; for the orthogonal instance the anchor-energy classifier of the main model realizes it (shared covariance, determinant factors cancel). Hence $L^* \le C_{\mathrm{G}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \le L \le L^* + \varepsilon \le C_{\mathrm{G}} + \varepsilon \le \ln K + \varepsilon$, whence the bound. ☐

Corollary 1.1 (Separation at near-optimal solutions, classification)**.** *Under the assumptions of Proposition 1.6, for classes $i \ne j$ with $\pi_i, \pi_j > 0$,*

$$\|m_i - m_j\| \;\ge\; \|m_\theta(i)-m_\theta(j)\| - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_i}} - \sqrt{\frac{2(C_{\mathrm{G}}+\varepsilon)}{\beta\,\pi_j}}, \tag{6}$$

which specializes via Δ. In particular, at a global optimum ($\varepsilon = 0$), classes with probability at least $\pi_{\min}$ have posterior centers separated by at least $\Delta - 2\sqrt{2C_{\mathrm{G}}/(\beta\,\pi_{\min})}$.

Proof. $D_k \le \mathbb{E}[D(Y)]/\pi_k \le (C_{\mathrm{G}}+\varepsilon)/(\beta\pi_k)$; combine with Proposition 1.5, whose mismatch terms are $\sqrt{2\tau_{\max}^2 D_k}$. ☐

Proposition 1.7 (Anchor alignment at near-optima: deterministic continuous regression)**.** *Assume the deterministic regression setting $Y = f(X)$ with a non-negative regression loss $\ell(\hat{y}, y)$ (e.g., the squared loss on standardized labels), a qualified family, a posterior able to realize the aligned solution $q(z|x) = p_\theta(z|y)$, and the tied projection head of the isometric instance. Let C_{align} be the regression risk of the ideal aligned solution — for the standardized squared loss and the EPB head, $\hat{\tilde{y}} = u^\top z/\rho$ with $z \sim p_\theta(z|y)$ gives $\hat{\tilde{y}} = \tilde{y} + (\tau/\rho)\,\eta$ with $\eta \sim \mathcal{N}(0,1)$, so $C_{\mathrm{align}} = \tau^2/\rho^2$. Then for any solution with objective value $L \le L^* + \varepsilon$,*

$$\mathbb{E}\big[D(Y)\big] \;\le\; \frac{C_{\mathrm{align}}+\varepsilon}{\beta}, \tag{7}$$

with $D(y)$ as above.

Proof. As in Proposition 1.6, $\mathbb{E}[\mathrm{KL}] \geq \mathbb{E}[D(Y)]$ by the covariance bound. The comparison solution is feasible by the assumptions ($Y = f(X)$, aligned maps representable): its posterior is $q(z|x) = p_\theta(z|y)$ (zero KL) and its prediction is the tied head, whose risk is $C_{\text{align}} = \tau^2/\rho^2$ (the prediction error is exactly the Gaussian bottleneck noise $(\tau/\rho)\,\eta$). Hence $L^* \leq C_{\text{align}}$ and $\beta\,\mathbb{E}[\mathrm{KL}] \leq L \leq L^* + \varepsilon \leq C_{\text{align}} + \varepsilon$, whence the bound. $\square$

Corollary 1.2 (Geometric guarantee on high-probability label sets, regression)**.** *Under Proposition 1.7, for any $t > 0$, Markov's inequality gives $P_Y\big(D(Y) > t\big) \leq (C_{\text{align}} + \varepsilon)/(\beta t)$, and on the label set $S_t := \{y : D(y) \leq t\}$, of P_Y-measure at least $1 - (C_{\text{align}} + \varepsilon)/(\beta t)$, the pointwise transfer of Proposition 1.5(b) yields, for all $y, y' \in S_t$,*

$$\|m(y) - m(y')\| \;\geq\; \rho_{\min}\, d_Y(y, y') - 2\sqrt{2\tau_{\max}^2\, t}. \tag{8}$$

For EPB ($\rho_{\min} = \rho$, $\tau_{\max} = \tau$) the bound reads $\rho\, d_Y(y, y') - 2\sqrt{2\tau^2 t}$: the posterior geometry is controlled by β, ρ, and the bottleneck noise τ — the optimization-level guarantee for the regression instance.

1.4 The orthogonal-frame instance

Proposition 1.8 (Orthogonal class-prior geometry by construction)**.** *Under the construction of the main text, the class means $a_k = a\, q_{\theta,k}$ satisfy*

$$a_i^\top a_j = a^2 \delta_{ij}, \qquad \|a_k\| = a, \qquad \|a_i - a_j\|^2 = 2a^2 \;\; (i \neq j). \tag{9}$$

Equivalently, the class-mean Gram matrix is $a^2 I_K$: the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.

Proof. The statement follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by a; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. $\square$

Proposition 1.9 (Positive rate cost of class-independent means)**.** *Let $\pi_k = \Pr(Y = k)$ and suppose the posterior mean is class-independent, $\mathbb{E}[\mu_\phi(X)|Y = k] = c$ for every k. Then the mean-mismatch part of the rate satisfies*

$$\mathbb{E}\Big[\tfrac{1}{2\tau^2}\|\mu_\phi(X) - a\, q_{\theta,Y}\|^2\Big] \;\geq\; \frac{a^2}{2\tau^2}\Big(1 - \sum_{k=1}^{K} \pi_k^2\Big), \tag{10}$$

with equality at the optimal constant mean $c = a\sum_k \pi_k q_{\theta,k}$. For uniform classes the bound is $a^2(1 - 1/K)/(2\tau^2)$. Unlike a zero-mean marginal prior, the geometric prior assigns every class-independent posterior a strictly positive mean-matching cost whenever $a > 0$ and two classes have positive probability.

Proof. By Jensen's inequality the left side is at least $\frac{1}{2\tau^2}\sum_k \pi_k\|c - aq_{\theta,k}\|^2 = \frac{1}{2\tau^2}\big(\|c\|^2 - 2a\, c^\top \sum_k \pi_k q_{\theta,k} + a^2\big)$. Minimizing over c gives $c = a\sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding the bound. $\square$

Proposition 1.10 (Anchor-energy classifier equivalence)**.** *Suppose the classifier uses the anchor energy $s_k(z) = -\frac{\|z-a\,q_{\theta,k}\|^2}{2\tau^2} + \log \pi_k$. Then $s_k(z)$ is the class-k Gaussian log posterior up to a class-independent term; equivalently, for uniform classes the equivalent linear classifier has weights $w_k = (a/\tau^2)\,q_{\theta,k}$ — equal norms and pairwise orthogonal rows. An anchor-energy classifier therefore makes the prediction rule use exactly the same geometry as the prior, and removes the free classifier scale by construction: its scale is fixed to a/τ^2, with no independent margin parameter.*

Proof. Expand the squared norm; the terms $-\|z\|^2/(2\tau^2)$ and $-a^2/(2\tau^2)$ are common to all classes, and the class-dependent remainder is the Gaussian discriminant score. □

2 Implementation details

2.1 Equivariance of the polar frame under class relabeling

For any label permutation matrix P,

$$Q_\theta(U_\theta P) = U_\theta P\big(P^\top U_\theta^\top U_\theta P\big)^{-1/2} = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2} P = Q_\theta(U_\theta)\,P, \qquad (11)$$

since $PP^\top = I_K$ and the inverse square root conjugates under P. Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

2.2 The full-rank condition

[Full rank] U_θ has full column rank at every training step, so that the polar factor — and its backward pass — is defined. The condition holds almost surely at the identity initialization; the implementation *verifies* it by monitoring $\sigma_{\min}(U_\theta)$ along training, and no additional regularizer is imposed.

2.3 What the framework constrains — and what it does not

Remark 2.1 (Scope). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y=k) = \mathcal{N}(a\,q'_k, \tau^2 I_d)$, the trainable frame can rotate to match ($q_{\theta,k} \to q'_k$), yielding KL $= 0$, $I(X;Z|Y) = 0$, and $I(Y;Z) > 0$ — a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code — precisely what CEB left unspecified — and does not claim that the certificate meters retained label information. (With the variance fixed, the remaining freedom at certificate zero is the frame rotation; a learnable per-class variance would add a further circumvention channel and is deliberately not used.)

2.4 Theory–code layering

The prior variance is the fixed shared value τ^2 (default 1.0) in both the theory and the released implementation: a learnable per-class variance is deliberately not used, as it would give the reference an extra channel to absorb the compression cost by inflating its covariance — the very degeneracy the geometric construction excludes. The released implementation uses exactly the polar prior encoder of the main text. Orthonormality holds exactly wherever the factorization is defined; the full-rank condition is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective. Near rank deficiency the polar factor's gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

2.5 Axis choices for EPB

Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \ldots, 0]$, with no trainable prior direction — the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with W trained and normalized at every step — our main variant, which keeps the scale ρ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$ — the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

2.6 Multi-dimensional regression labels

For multi-dimensional regression labels $y \in \mathbb{R}^m$ ($m \leq d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d\times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho\, Q\, \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise "isometric" has no unique meaning.

3 Deferred results

Lemma 3.1 (OPB KL decomposition)**.** *For a diagonal posterior covariance and the OPB prior,*

$$\mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big) = \frac{1}{2\tau^2}\|\mu_\phi(x) - a\, q_{\theta,y}\|^2 + \frac{1}{2}\sum_{r=1}^{d}\Big[\frac{\sigma^2_{\phi,r}(x)}{\tau^2} - 1 - \log\frac{\sigma^2_{\phi,r}(x)}{\tau^2}\Big], \tag{12}$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma^2_{\phi,r}(x) = \tau^2$ for all r. The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition 3.1 below.

Proof. Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a\, q_{\theta,y}$, posterior variances $\sigma^2_{\phi,r}(x)$, and the fixed prior variance τ^2. □

Corollary 3.1 (Label-information lower bound in the aligned Gaussian model)**.** *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a\, q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\mathrm{pair}} = \Phi\big(-\frac{a}{\sqrt{2}\,\tau}\big)$, with $P_e \le \min\{1, (K-1)P_{\mathrm{pair}}\}$, and whenever the displayed bound $\bar{P}_e \le 1-1/K$, Fano's inequality gives $I(Y;Z) \ge \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1)$. This links the anchor scale-to-noise ratio a/τ to the retained label information, conditionally on the aligned Gaussian model.*

Proof. Two anchors have distance $\sqrt{2}a$; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano's inequality on $[0, 1 - 1/K]$ conclude. $\square$

Proposition 3.1 (Joint strict convexity and unique optimum on a restricted aligned surrogate)**.** *Consider the class-symmetric aligned family: every input of class y is mapped to the posterior $q_y = \mathcal{N}(s\, q_{\theta,y}, \sigma^2 I)$ ($s \ge 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier (weights $w_j = c\, q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the* stochastic training code *$z \sim q_y$ — as in the actual objective — the training cross-entropy is*

$$g(s, \sigma^2) := \mathbb{E}_\eta\Big[\ln\Big(1 + \textstyle\sum_{j \ne y} e^{-cs + c\sigma(\eta_j - \eta_y)}\Big)\Big], \qquad \eta \sim \mathcal{N}(0, I_K), \tag{13}$$

and the training Lagrangian is

$$L_\beta(s, \sigma^2) = g(s, \sigma^2) + \beta\Big[\tfrac{1}{2\tau^2}(s - a)^2 + \tfrac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)\Big]. \tag{14}$$

Then for every $\beta > 0$, L_β is strictly convex jointly in (s, σ) and has a unique joint minimizer $(s^*(\beta), \sigma^*(\beta))$, interior in s since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every β. This is a restricted surrogate result*: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective — with its nonlinear posterior, trainable frame, and non-convex degrees of freedom — lies outside this analysis, and the result neither establishes β-tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ($z = \mu$) is the $\sigma \to 0$ limit, and $g \ge \ln(1+(K-1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do* not *claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer σ^* responds to β through the cross-derivative $\partial^2 g/\partial s\, \partial\sigma$, and establishing the sign of $\mathrm{d}s^*/\mathrm{d}\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the β-sweep is reported empirically. The fixed-scale condition holds for the main model by construction, so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution — at $\mathrm{KL} \to 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$*

it contains $\mathbb{E}[\ln(1 + e^{-c(a+\tau\xi)})]$*, whose integrand grows linearly in* c *on the event* $\xi < -a/\tau$*, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.*

Proof. For each fixed η, the integrand is a log-sum-exp of functions affine in (s, σ) (the constant 1 has slope zero and each exponential has slope $(-c,\, c(\eta_j - \eta_y))$), hence jointly convex in (s, σ), and the expectation g is jointly convex. No strictness is claimed for g: for $K = 2$ the integrand depends on (s, σ) only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s, \sigma) := \frac{1}{2\tau^2}(s - a)^2 + \frac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)$; its Hessian in (s, σ) is $\nabla^2 R = \operatorname{diag}\big(\tau^{-2},\ d\,(\tau^{-2} + \sigma^{-2})\big)$, strictly positive definite for every $\sigma > 0$, so βR is strictly convex in (s, σ). Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in (s, σ). Since $R \to +\infty$ as $s \to \infty$, $\sigma \to 0^+$, or $\sigma \to \infty$, the sublevel sets of L_β are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in s: at $s = 0$, $\partial L_\beta/\partial s = -c\,\mathbb{E}[p_{\mathrm{w}}] - \beta a/\tau^2 < 0$, where p_{w} is the error probability of the sampled code, so $s^* > 0$. □

Remark 3.1 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition 3.1 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale — a restricted surrogate result, which does not establish β-tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB's — the certificate remains the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on the full-rank assumption: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified — not imposed — along the reported runs; the construction is exactly equivariant under relabeling of the classes, so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and the certificate-preservation argument would no longer follow directly. Finally, the anchor-energy classifier fixes the classifier scale to a/τ^2 by construction, removing the independent classifier-margin escape route in the aligned surrogate; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition 3.1 moves monotonically in β is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to β through the cross-derivative $\partial^2 g/\partial s\,\partial\sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate's zero set contains the full-retention orthogonal label codes at any frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code* — it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

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